Oliver's Multivariable Calculus Cram Study Guide

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Contents

Chapter 1

Preliminaries

A **proof** is a series of logical statements that essentially shows that a given claim is either true or false. Proofs should convince the reader why a claim is true as well. We use the symbols $\forall$ to say "for all" and $\exists$ to say "there exists."

To negate statements, we follow the following rules:

1. If we have a statement $\forall x, P(x)$ is true, the opposite is $\exists x$ such that $P(x)$ is not true

2. If we have a statement $\exists x$, such that $R(x)$ is true, the opposite is $\forall x$, $R(x)$ is not true

We need to understand some set theory. A **set** is a *well-defined* collection of elements. The symbol $\in$ means "is an element of." If we have an expression of the form

$$\{a|p(a)\} \tag{1.1}$$

then we are considering the set of all a such that $p(a)$ is true. The symbol $\subset$ means "is a **subset** of, or equals." Note that $A \subset A$ for any set A. The **intersection** is denoted by the symbol $\cap$; $A \cap B$ is the set of all elements that are in both A and B. The **union** is denoted by the symbol $\cup$: $A \cup B$ is the set of all elements that are in either A, B, or both. The **Cartesian product** is denoted by the times symbol, $\times$. We have that $A \times B$ is the set of all ordered pairs (a, b), where $a \in A$ and $b \in B$. The **complement**, or difference, is denoted by the minus sign, $-$. $A - B$ is the set of all elements that are in A but not in B.

The empty set, which has no elements, is denoted by $\emptyset = \{\}$. We also have sets of numbers that are denoted in blackboard bold, like so:

- $\mathbb{N}$ is the set of **natural numbers**: $\{1, 2, 3, \ldots\}$ (0 may or may not be a natural number, depending on your source. Hubbard and Hubbard says it is, I say it isn't.)
- $\mathbb{Z}$ is the set of **integers**: $\{\ldots, -1, 0, 1, \ldots\}$
- $\mathbb{Q}$ is the set of **rational numbers**, which are numbers of the form p/q, where $p, q \in \mathbb{Z}$ and $q \neq 0$
- $\mathbb{R}$ is the set of **real numbers**, which can be constructed rigorously; you should know what they are "unrigorously," that is, rational and irrational numbers
- $\mathbb{C}$ is the set of **complex numbers**, which are represented by the form $a + bi$, where $a, b \in \mathbb{R}$ and $i^2 = -1$

One should note that

$$\bigcap_{\alpha \in A} S_\alpha \tag{1.2}$$

is the intersection of all sets S_α, where $\alpha \in A$. We have similar notation for the union of multiple sets.

Now, we say that a number a is an **upper bound** of a subset $X \subset \mathbb{R}$ if $\forall x \in X$, then $x \leq a$. The **least upper bound**, or **supremum**, is an upper bound b such that $\forall a$ that is an upper bound, $b \leq a$. We say $b = \sup X$. We consider the following theorem related to least upper bounds:

Theorem 1. *If some non-empty subset $X \subset \mathbb{R}$ has an upper bound, then it also has a least upper bound,* $\sup X$.

Proof. Suppose there exists some $x \in X$, and a is an upper bound of X. Assume $x > 0$ for this proof. If $x = a$, then a is obviously the least upper bound. If $x \neq a$, then there must be some j such that the jth (decimal) digit of x must be smaller than that of a. We now repeat the following process iteratively: consider the set $[x, a]$ with numbers written using j digits only after the decimal. Suppose b_j is the largest which is not an upper bound. Then consider the set $[b_j, a]$ as before, and select a b_{j+1} as before. Repeat the process iteratively. Suppose b is the number whose kth decimal digit is the same as that of b_k; this is $b = \sup X$. It is an easy exercise for the reader to show that this fact; consider any $y \in X$ such that $y > b$, if such a number can exist, and consider $b' < b$ where b' is an upper bound, if that number even exists. In fact, neither exist, so $b = \sup X$, and therefore $\sup X$ exists given the conditions we have stated. $\square$

Suppose that $\mathbb{D}$ represents the set of finite decimals (that is, decimals that do not go on forever). We define a mapping (function) $f : \mathbb{D}^n \to \mathbb{D}$ (takes in n finite decimal inputs and spits out one finite decimal output) to be **finite decimal continuous**, or $\mathbb{D}$-continuous, if $\forall N, k \in \mathbb{Z}$, $\exists l$ such that if $(x_1, ..., x_n)$ and $(y_1, ..., y_n) \in \mathbb{D}^n$, such that $|x_i|, |y_i| < N$ and $|x_i - y_i| < 10^{-l}$ $\forall i = 1, ..., n$, then we must have

$$|f(x_1, ..., x_n) - f(y_1, ..., y_n)| < 10^{-k} \tag{1.3}$$

Note the similarities between this and the single-variable epsilon-delta definition of a limit. We will soon see the multivariable epsilon-delta definition, which possesses striking similarities to this definition.

Now, we say that two points $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are k-close if $\forall i = 0, ..., n$, then we have $|[x_i]_k - [y_i]_k| \leq 10^{-k}$, where $[x_i]_k$ is the truncation of x_i to k decimals.

We can say that there is a unique function $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ such that $\tilde{f}(\mathbf{x}) = \sup_k \inf_{l \geq k} f([x_1]_l, ..., [x_n]_l)$ coinciding with a function f on $\mathbb{D}^n$ and $\forall k \in \mathbb{N} \forall N \in \mathbb{N} \exists l \in \mathbb{N}$ such that $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are l-close and $|x_i| < N$ for all i implies that $\tilde{f}(\mathbf{x})$ and $\tilde{f}(\mathbf{y})$ are k-close.

We consider the definition of a **convergent sequence**. A sequence a_n of real numbers converges to a limit a if $\forall \epsilon > 0$, there exists N such that $\forall n > N$, $|a - a_n| < \epsilon$.

Meanwhile, we say that if the sequence of *partial sums* of a series converges to a value S, then the series converges, and we have

$$\sum_{n=1}^{\infty} a_n = S \tag{1.4}$$

For example, if we have some $|r| < 1$, then

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r} \tag{1.5}$$

The derivation of this equation is a standard exercise in basic algebra.

We state the following theorem:

Theorem 2. *A non-decreasing sequence a_n must converge if and only if it is bounded.*

Proof. We see the "if and only if," so we must prove in both directions. Assume a_n is bounded and non-decreasing. Then it must have a least upper bound A. Choose an $\epsilon > 0$; suppose for the sake of contradiction that $A - a_n > \epsilon$ for all n; then $A - \epsilon > a_n$ and $A - \epsilon$ would be an upper bound, contradicting the fact that A is the least upper bound. Therefore, there must be some N such that

$A - a_N < \epsilon$, and furthermore, $A - a_n < A - a_N < \epsilon$ for all $n \geq N$. Now, for the other direction, suppose a_n converges to a value A. Since a_n is non-decreasing, then $a_n \leq A$ for all a_n, and therefore it is easy to see that the sequence a_n is bounded. □

We also have that:

Theorem 3. *If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges as well.*

Proof. Consider the series

$$\sum_{n=1}^{\infty}(a_n + |a_n|) \tag{1.6}$$

where $a_n + |a_n|$ is non-negative, as one can easily verify, which implies that the partial sums are non-decreasing. The partial sums are also bounded as well, as

$$\sum_{n=1}^{m}(a_n + |a_n|) \leq \sum_{n=1}^{m} 2|a_n| = 2\sum_{n=1}^{m} |a_n| \leq 2\sum_{n=1}^{\infty} |a_n| \tag{1.7}$$

where the last expression converges to a certain value by our givens. Since the first series is both non-decreasing and bounded, then it must converge, and we have that

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty}(a_n + |a_n|) - \sum_{n=1}^{\infty} |a_n| \tag{1.8}$$

is the sum (or difference) of two convergent series, and therefore must be convergent as well. □

There is also a very important theorem that you may remember from single-variable calculus, known as the intermediate value theorem:

Theorem 4. *If we have a continuous $f : [a, b] \to \mathbb{R}$ such that $f(a) \leq 0$ and $f(b) \geq 0$, then there exists some $c \in [a, b]$ such that $f(c) = 0$.*

Proof. Note that this is just a special case of the actual intermediate value theorem. Suppose $X \subset [a, b]$ such that $f(x) \leq 0$ for all $x \in X$. X is clearly non-empty, and has an upper bound (b), so it must have a least upper bound, which we call c. We need to show that $f(c) = 0$. By the definition of continuity, we have that

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } |x - c| < \delta \implies |f(x) - f(c)| < \epsilon \tag{1.9}$$

Assume for the sake of contradiction that $f(c) > 0$. Letting $\epsilon = f(c)$, there must exist some $\delta > 0$ such that

$$|x - c| < \delta \implies |f(x) - f(c)| < f(c) \tag{1.10}$$

Suppose $x > c - \delta/2$. Then we must have that $|x - c| < \delta$, which implies that $|f(x) - f(c)| < f(c)$. Since $f(x) \leq 0$ as stated before and $f(c) > 0$, it follows that $f(x) - f(c) < 0$, and so

$$-f(x) + f(c) < f(c) \implies f(x) > 0 \tag{1.11}$$

This is a contradiction. This implies that $c - \delta/2$ is an upper bound for X, whereas c is supposed to be the least upper bound for X. Therefore, we cannot have that $f(c) > 0$. By a similar argument, we cannot have that $f(c) < 0$, either. Therefore, we must have that $f(c) = 0$. As c is guaranteed to exist, then there exists some $c \in [a, b]$ such that $f(c) = 0$. $\square$

Chapter 2

Vectors

What is a vector? A **vector** is considered to be an element of a vector space, but for the purposes of right now, we consider a vector to be some quantity that contains both magnitude and direction. Vectors like these can be represented by **directed line segments**, in which there exists an initial point and a terminal point. The **standard form** of the vector is the representation in which the vector's initial point is at the origin. Furthermore, we can use coordinates to split a vector up into **components**, in which the component form of a vector $\mathbf{v}$ *in a plane* is written like

$$\langle v_1, v_2 \rangle \tag{2.1}$$

where v_1 and v_2 are the components of the terminal point (the initial point is the origin when we are considering component form). The **zero vector** is the vector $\mathbf{0} = \langle 0, 0 \rangle$.

Note that we can also write n-dimensional vectors in the form

$$\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \quad \text{or} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \tag{2.2}$$

The angle bracket notation is usually not used in more advanced texts.

We have that two vectors $\langle u_1, u_2 \rangle$ and $\langle v_1, v_2 \rangle$ are equal if $u_1 = v_1$ and $u_2 = v_2$. The magnitude, or length, of the vector $\mathbf{v}$ is

$$||\mathbf{v}|| = \sqrt{v_1^2 + v_2^2} \tag{2.3}$$

Sooner or later we will see that this magnitude can be generalized to something called the norm. If $||\mathbf{v}|| = 1$, then $\mathbf{v}$ is a unit vector. It is also clear that $||\mathbf{v}|| = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

We shall define vector addition and scalar multiplication componentwise. Given $\mathbf{u} = \langle u_1, u_2 \rangle$ and $\mathbf{v} = \langle v_1, v_2 \rangle$, we have:

$$\mathbf{u} + \mathbf{v} = \langle u_1 + v_1, u_2 + v_2 \rangle \tag{2.4}$$

$$c\mathbf{u} = \langle cu_1, cu_2 \rangle \tag{2.5}$$

$$-\mathbf{v} = (-1)\mathbf{v} \tag{2.6}$$

$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v}) \tag{2.7}$$

The quantities above are labeled the **vector sum**, the **scalar multiple**, the **negative**, and the **difference**, respectively. Recall that a **scalar** is a value that can be represented by a single number.

Consider the following properties of vector operations:

Theorem 5. *Suppose* $\boldsymbol{u}$*,* $\boldsymbol{v}$*, and* $\boldsymbol{w}$ *are vectors in the plane, and* c *and* d *are scalars. We have that*

1. *Vector addition is commutative:* $\boldsymbol{u}+\boldsymbol{v}=\boldsymbol{v}+\boldsymbol{u}$

2. *Vector addition is associative:* $(\boldsymbol{u}+\boldsymbol{v})+\boldsymbol{w}=\boldsymbol{u}+(\boldsymbol{v}+\boldsymbol{w})$

3. *There exists an additive identity* $\boldsymbol{0}$ *such that* $\boldsymbol{u}+\boldsymbol{0}=\boldsymbol{u}$

4. *There exists, for each vector* $\boldsymbol{u}$*, an additive inverse* $-\boldsymbol{u}$ *such that* $\boldsymbol{u}+(-\boldsymbol{u})=0$

5. *Scalar multiplication follows the following property:* $c(d\boldsymbol{u})=(cd)\boldsymbol{u}$

6. *There exist two distributive properties for vectors:* $(c+d)\boldsymbol{u}=c\boldsymbol{u}+d\boldsymbol{u}$

7. *The second distributive property:* $c(\boldsymbol{u}+\boldsymbol{v})=c\boldsymbol{u}+c\boldsymbol{v}$

8. *There exists a multiplicative identity* 1 *such that* $1\boldsymbol{u}=\boldsymbol{u}$*, and the additive identity* 0 *follows the property* $0\boldsymbol{u}=\boldsymbol{0}$

The proof for these properties is fairly straightforward, and can be accomplished by considering the properties of the real numbers.

We consider the length of a scalar multiple of a vector, and we find that it has a formula:

Theorem 6. *Suppose* $\boldsymbol{v}$ *is a vector and* c *is a scalar. Then the following formula holds:*

$$||c\boldsymbol{v}||=|c|||\boldsymbol{v}|| \tag{2.8}$$

Proof. The proof for this theorem involves a few lines of equations. Let $\mathbf{v}=\langle v_1,v_2\rangle$. By the definition of the magnitude,

$$||c\mathbf{v}||=||\langle cv_1,cv_2\rangle||=\sqrt{(cv_1)^2+(cv_2)^2} \tag{2.9}$$

$$= \sqrt{c^2v_1^2 + c^2v_2^2} = \sqrt{c^2(v_1^2 + v_2^2)} \tag{2.10}$$

$$= \sqrt{c^2}\sqrt{v_1^2 + v_2^2} = |c|||\mathbf{v}|| \tag{2.11}$$

□

For any nonzero vector, we can create a unit vector in the direction of the vector like so:

Theorem 7. *The vector*

$$\boldsymbol{u} = \frac{\boldsymbol{v}}{||\boldsymbol{v}||} \tag{2.12}$$

must have magnitude 1 and must point in the same direction as $\boldsymbol{v}$*.*

Proof. Because $\mathbf{u}$ is a scalar multiple of $\mathbf{v}$, and because $1/||\mathbf{v}||$ is positive, we must have that $\mathbf{u}$ points in the same direction of $\mathbf{v}$. For the other claim, we have that

$$||\mathbf{u}|| = \left|\left|\left(\frac{1}{||\mathbf{v}||}\right)\mathbf{v}\right|\right| \tag{2.13}$$

$$= \left|\frac{1}{||\mathbf{v}||}\right| ||\mathbf{v}|| \tag{2.14}$$

$$= \frac{1}{||\mathbf{v}||}||\mathbf{v}|| = 1 \tag{2.15}$$

Therefore, $\mathbf{u}$ must also have magnitude 1. □

Furthermore, we define $\mathbf{i}$ and $\mathbf{j}$ to be the **standard unit vectors**, where $\mathbf{i} = \langle 1, 0\rangle$ and $\mathbf{j} = \langle 0, 1\rangle$.

In three-dimensional space, there are a couple important formulas. The **distance formula** is undoubtedly familiar; given two points (x_1, x_2, x_3) and (y_1, y_2, y_3), the distance between them is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \tag{2.16}$$

The standard equation for a **sphere**, in which all points (x, y, z) on the sphere are some r distance away from a point (x_0, y_0, z_0), is

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = r^2 \tag{2.17}$$

The **midpoint formula** must also be familiar; given two points (x_1, y_1, z_1) and (x_2, y_2, z_2), their midpoint is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right) \tag{2.18}$$

The **standard unit vectors** $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are defined analogously to those in a plane, where $\mathbf{i} = \langle 1, 0, 0\rangle$, and so on. Vectors in space follow certain properties:

- $\mathbf{u} = \mathbf{v}$ if and only if $u_1 = v_1$, $u_2 = v_2$, and $u_3 = v_3$.
- $||\mathbf{v}|| = \sqrt{v_1^2 + v_2^2 + v_3^2}$
- $\frac{\mathbf{v}}{||\mathbf{v}||} = \frac{1}{||\mathbf{v}||}\langle v_1, v_2, v_3\rangle$, if $\mathbf{v} \neq \mathbf{0}$
- $\mathbf{v} + \mathbf{u} = \langle v_1 + u_1, v_2 + u_2, v_3 + u_3\rangle$
- $c\mathbf{v} = \langle cv_1, cv_2, cv_3\rangle$

Now, we say that two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are **parallel** if we have that there exists some c such that $\mathbf{u} = c\mathbf{v}$.

The **dot product** of two plane vectors $\mathbf{u}$ and $\mathbf{v}$ is

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 \tag{2.19}$$

For space vectors, we have

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3 \tag{2.20}$$

Finally, for vectors in general, we have

$$\vec{\mathbf{x}} \cdot \vec{\mathbf{y}} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = x_1y_1 + x_2y_2 + \cdots + x_ny_n \tag{2.21}$$

It should be noted that we have that

$$\|\vec{\mathbf{x}}\| = \sqrt{\vec{\mathbf{x}} \cdot \vec{\mathbf{x}}} \tag{2.22}$$

The dot product has some properties of note:

Theorem 8. *If $\boldsymbol{u}$, $\boldsymbol{v}$, and $\boldsymbol{w}$ are vectors, and c is a scalar,*

1. $\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{u}$
2. $\boldsymbol{u} \cdot (\boldsymbol{v} + \boldsymbol{w}) = \boldsymbol{u} \cdot \boldsymbol{v} + \boldsymbol{u} \cdot \boldsymbol{w}$
3. $c(\boldsymbol{u} \cdot \boldsymbol{v}) = c\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{u} \cdot c\boldsymbol{v}$
4. $\boldsymbol{0} \cdot \boldsymbol{v} = 0$
5. $\boldsymbol{v} \cdot \boldsymbol{v} = ||\boldsymbol{v}||^2$

Once more, the proof can be found using the properties of real numbers, or, in the case of the fifth property, by considering the definition of the magnitude.

We note that the **angle** between two vectors **u** and **v** can be described by the formula

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||||\mathbf{v}||} \tag{2.23}$$

It follows shortly that the dot product can also be defined as

$$\mathbf{u} \cdot \mathbf{v} = ||\mathbf{u}||||\mathbf{v}|| \cos\theta \tag{2.24}$$

where θ is the angle between the two vectors.

We define two vectors $\mathbf{u}$ and $\mathbf{v}$ to be **orthogonal** if $\mathbf{u} \cdot \mathbf{v} = 0$.

The **projection** of a vector $\mathbf{u}$ onto a vector $\mathbf{v}$ can be described by the formula

$$\text{proj}_{\mathbf{v}}\mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||^2}\right)\mathbf{v} \tag{2.25}$$

The scalar

$$\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{v}||} \tag{2.26}$$

is called the **component** of $\mathbf{u}$ in the direction of $\mathbf{v}$. No, the power of the magnitude in the denominator is not a typo; it is *not* squared, because $\mathbf{v}/||\mathbf{v}||$ is the unit vector pointing in the direction of $\mathbf{v}$, so the scalar multiplying that in the formula for the projection should be the component.

We now come to a famous inequality, known as **Schwarz's Inequality**:

Theorem 9. *For any vectors $\vec{\boldsymbol{v}}, \vec{\boldsymbol{w}}$, we have that*

$$|\vec{\boldsymbol{v}} \cdot \vec{\boldsymbol{w}}| \leq \|\vec{\boldsymbol{v}}\|\|\vec{\boldsymbol{w}}\| \tag{2.27}$$

Proof. Consider the function $f(t) = \|\vec{\mathbf{v}}+t\vec{\mathbf{w}}\|^2$. It is a second-degree polynomial of t, which we can see clearly below:

$$\|\vec{\mathbf{v}} + t\vec{\mathbf{w}}\|^2 = \|\vec{\mathbf{w}}\|^2t^2 + 2(\vec{\mathbf{v}} \cdot \vec{\mathbf{w}})t + \|\vec{\mathbf{v}}\|^2 \tag{2.28}$$

Recall a key property of the discriminant of a quadratic, $b^2 - 4ac$: if the discriminant is positive, then the equation set to zero has two distinct solutions, and therefore the graph should cross the t-axis twice (and so will have negative solutions as well). This cannot be the case, as $f(t)$ can only take non negative solutions (the square of a number must be non negative). The discriminant of our equation is

$$4(\vec{\mathbf{v}} \cdot \vec{\mathbf{w}})^2 - 4\|\vec{\mathbf{v}}\|^2\|\vec{\mathbf{w}}\|^2 \tag{2.29}$$

as $b = 2(\vec{\mathbf{v}} \cdot \vec{\mathbf{w}})$ and $a = \|\vec{\mathbf{w}}\|^2$ and $c = \|\vec{\mathbf{v}}\|^2$. This discriminant must be non positive, so

$$4(\vec{\mathbf{v}} \cdot \vec{\mathbf{w}})^2 - 4\|\vec{\mathbf{v}}\|^2\|\vec{\mathbf{w}}\|^2 \leq 0 \implies |\vec{\mathbf{v}} \cdot \vec{\mathbf{w}}| \leq \|\vec{\mathbf{v}}\|\|\vec{\mathbf{w}}\| \tag{2.30}$$

□

Another useful inequality is the **triangle inequality**,

Theorem 10. *For any vectors $\vec{\boldsymbol{x}}, \vec{\boldsymbol{y}} \in \mathbb{R}^n$, we have that*

$$\|\vec{\boldsymbol{x}} + \vec{\boldsymbol{y}}\| \leq \|\vec{\boldsymbol{x}}\| + \|\vec{\boldsymbol{y}}\| \tag{2.31}$$

Proof. We have

$$\|\vec{\mathbf{x}} + \vec{\mathbf{y}}\|^2 \leq \|\vec{\mathbf{x}}\|^2 + 2\vec{\mathbf{x}} \cdot \vec{\mathbf{y}} + \|\vec{\mathbf{y}}\|^2 \tag{2.32}$$

which, by the Schwarz inequality, is less than or equal to

$$\|\vec{\mathbf{x}}\|^2 + 2\|\vec{\mathbf{x}}\|\|\vec{\mathbf{y}}\| + \|\vec{\mathbf{y}}\|^2 = (\|\vec{\mathbf{x}}\| + \|\vec{\mathbf{y}}\|)^2 \tag{2.33}$$

Hence, $\|\vec{\mathbf{x}} + \vec{\mathbf{y}}\| \leq \|\vec{\mathbf{x}}\| + \|\vec{\mathbf{y}}\|$ □

The **cross product**, or vector product, of two vectors $\mathbf{u}$ and $\mathbf{v}$ renders a perpendicular vector

$$\mathbf{u} \times \mathbf{v} = (u_2v_3 - u_3v_2)\mathbf{i} - (u_1v_3 - u_3v_1)\mathbf{j} + (u_1v_2 - u_2v_1)\mathbf{k} \tag{2.34}$$

Note that the cross product only exists in 3 dimensions and... oddly enough, 7 dimensions as well? The cross product can also be represented in terms of the

determinant, if you are familiar with that:

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} \tag{2.35}$$

The cross product exhibits certain algebraic properties. For instance:

Theorem 11. *We have that for vectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ and scalar c,*

1. *The cross product is not commutative; in fact, it is* ***anti-commutative****:* $\boldsymbol{u} \times \boldsymbol{v} = -(\boldsymbol{v} \times \boldsymbol{u})$

2. $\boldsymbol{u} \times (\boldsymbol{v} + \boldsymbol{w}) = (\boldsymbol{u} \times \boldsymbol{v}) + (\boldsymbol{u} \times \boldsymbol{w})$

3. $c(\boldsymbol{u} \times \boldsymbol{v}) = (c\boldsymbol{u}) \times \boldsymbol{v} = \boldsymbol{u} \times (c\boldsymbol{v})$

4. $\boldsymbol{u} \times \boldsymbol{0} = \boldsymbol{0} \times \boldsymbol{u} = \boldsymbol{0}$

5. $\boldsymbol{u} \times \boldsymbol{u} = \boldsymbol{0}$

6. $\boldsymbol{u} \cdot (\boldsymbol{v} \times \boldsymbol{w}) = (\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w}$

The proofs for all of these properties are tedious, but can be done rather easily with some algebraic manipulation.

Furthermore, note that $||\mathbf{u} \times \mathbf{v}|| = ||\mathbf{u}|| ||\mathbf{v}|| \sin\theta$, where θ is the angle between the two vectors. We also have that $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ if and only if the two vectors are scalar multiples of each other. Finally, the cross product has an interesting geometric property in that the magnitude of the cross product is the area of the parallelogram having the two vectors as adjacent sides.

The **triple scalar product** is defined as

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} \tag{2.36}$$

$$= u_1 v_2 w_3 + u_2 v_3 w_1 + u_3 v_1 w_2 - u_3 v_2 w_1 - u_2 v_1 w_3 - u_1 v_3 w_2 \tag{2.37}$$

The triple scalar product has a geometric property, in that the volume V of a parallepiped with the three vectors as adjacent edges is precisely given by the absolute value of the triple scalar product of the three vectors.

Let us consider lines in space. They can be represented by the parametric equations

$$x = x_1 + at, \quad y = y_1 + bt, \quad z = z_1 + ct \tag{2.38}$$

Solving for t for all three of the parametric equations renders the **symmetric equations**:

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} \tag{2.39}$$

For a plane in space, the **standard form**, if the plane contains the point (x_1, y_1, z_1) and has a vector $\mathbf{n} = \langle a, b, c \rangle$ normal to it, is

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0 \tag{2.40}$$

This makes intuitive sense, if we consider that the dot product between the normal vector and the distance vector between some point on the plane and (x_1, y_1, z_1) must be 0, since they should be orthogonal to each other. The **general form** of the plane can be described similarly:

$$ax + by + cz + d = 0 \tag{2.41}$$

for some d.

Note that the angle between the two planes is equal to the angle between their normal vectors, so if we have normal vectors $\mathbf{n}_1, \mathbf{n}_2$ to some planes with angle θ, then

$$\cos\theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{||\mathbf{n}_1||||\mathbf{n}_2||} \tag{2.42}$$

To find the distance between a plane and a point Q note in the plane, then

$$D = \frac{|\vec{PQ} \cdot \mathbf{n}|}{||\mathbf{n}||} \tag{2.43}$$

where $\vec{PQ}$ is the vector from some point P in the plane to Q and $\mathbf{n}$ is normal to the plane.

One can verify therefore that the distance between the point (x_0, y_0, z_0) and the plane given by the equation $ax + by + cz + d = 0$ is

$$D = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}} \tag{2.44}$$

Furthermore, the distance between a point Q and a line in space is given by

$$D = \frac{||\vec{PQ} \times \mathbf{u}||}{||\mathbf{u}||} \tag{2.45}$$

where $\mathbf{u}$ is a vector pointing in the direction of the line and P is a point on the line.

Suppose we have a curve C and a line L not in a parallel plane. The set of all lines parallel to L and intersecting C is called a **cylinder**, and C is called the **directix** of the cylinder, and the parallel lines are called **rulings**.

If we have a cylinder parallel to one of the coordinate axes, it must only contain the variables corresponding to the other two axes. For example, for the equation $z = y^2$, the rulings of the cylinder are parallel to the x-axis.

We now consider quadric surfaces, whose equations take the form

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0 \tag{2.46}$$

It is simply a second-degree equation with any or all of the three variables. There are a couple kinds of quadric surfaces that must be known. The ellipsoid takes the form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \tag{2.47}$$

The hyperboloid of one sheet is described by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \tag{2.48}$$

The important thing is that there all terms contain the square of a variable, and one of those terms is negative; it doesn't matter if it's z^2 that's negative or x^2 or whatever. The hyperboloid of two sheets is described by

$$\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \tag{2.49}$$

A similar sentiment holds here as above. The elliptic cone has the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0 \tag{2.50}$$

The elliptic paraboloid has the equation

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2} \tag{2.51}$$

and last but not least, the hyperbolic paraboloid takes the form

$$z = \frac{y^2}{b^2} - \frac{x^2}{a^2} \tag{2.52}$$

To distinguish between them may be difficult. But fear not, for I have prepared a guide for you to distinguish the basic quadric surfaces. Read this like a choose-your-own-adventure:

1. Does the equation have a term with degree 1 (i.e. x, y, z), or is it all just terms that contain squares of variables (i.e. x^2, y^2, z^2)? If it is the former, go to item 2. If it is the latter, go to item 3.

2. When you rewrite the equation so that the variable of degree 1 is isolated, do you have that the signs for the other two terms are different (that is, positive and negative)? Or are they both positive or both negative? If it is the former, you have a **hyperbolic paraboloid**. If it is the latter, you have an **elliptic paraboloid**.

3. When you write the terms of the form x^2/a^2 on one side and factor so 1 or 0 is on the other side, do you have one term that is negative whereas the rest are positive? If so, go to item 4. If not, go to item 5.

4. When you do so, do you have a number that is nonzero on the other side (probably 1), or do you have a number that is zero on the other side? If it is the former, you have a **hyperboloid of one sheet**. If it is the latter, you have an **elliptic cone**.

5. Are all terms positive, or are two terms negative and one term positive? If it is the former, you have an **ellipsoid**. If it is the latter, you have a **hyperboloid of two sheets**. If all terms are somehow negative, you do not have a valid equation, for the sum of negative numbers cannot equal 1.

Maybe you remember a **surface of revolution** from Calculus BC. If you do, great! But here's a recap: the equation of a surface of revolution can be described based on the following:

1. If the surface revolves around the x-axis, the equation is $y^2 + z^2 = [r(x)]^2$

2. If the surface revolves around the y-axis, the equation is $x^2 + z^2 = [r(y)]^2$

3. If the surface revolves around the z-axis, the equation is $x^2 + y^2 = [r(z)]^2$

Note that $r(x), r(y), r(z)$ is called the **radius function**.

Let's talk about coordinate systems. The **cylindrical coordinate system** is described by an ordered triple (r, θ, z) where (r, θ) is a polar representation of the projection of the point onto the xy-plane, and z is the distance from $(r, \theta, 0)$ to the point. We have a couple equations that allow us to convert between cylindrical and rectangular coordinate systems:

$$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z \tag{2.53}$$

$$r^2 = x^2 + y^2, \quad \tan\theta = \frac{y}{x}, \quad z = z \tag{2.54}$$

For **spherical coordinates**, we have an ordered triple (ρ, θ, ϕ), where ρ is the distance between the point and the origin (it must be greater than or equal to zero!), θ is the same in cylindrical coordinates, and ϕ is the angle between the positive z-axis and the line segment from the origin to the point, it must be between 0 and π. To convert between spherical and rectangular, we have:

$$x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta, \quad z = \rho\cos\phi \tag{2.55}$$

$$\rho^2 = x^2 + y^2 + z^2, \quad \tan\theta = \frac{y}{x}, \quad \phi = \arccos\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right) \tag{2.56}$$

To convert between spherical and cylindrical, we have:

$$r^2 = \rho^2\sin^2\phi, \quad \theta = \theta, \quad z = \rho\cos\phi \tag{2.57}$$

$$\rho = \sqrt{r^2 + z^2}, \quad \theta = \theta, \quad \phi = \arccos\left(\frac{z}{\sqrt{r^2+z^2}}\right) \tag{2.58}$$

Fun fact: in physics, the notation convention is that θ and ϕ are swapped; that is, ϕ is the horizontal angle, and θ is the angle from the positive z-axis. Don't get confused in upper level physics courses!

Suppose that we have a subset $V \subset \mathbb{R}^n$, where $\mathbb{R}^n$ means the set of ordered n-tuples of real numbers. V is a **subspace** if when $\vec{\mathbf{x}}, \vec{\mathbf{y}} \in V$ and $a \in \mathbb{R}$, then $\vec{\mathbf{x}} + \vec{\mathbf{y}} \in V$ and $a\vec{\mathbf{x}} \in V$.

$\mathbb{R}^n$ is also equipped with **standard basis vectors** $\vec{\mathbf{e}}_j$, where the jth entry is 1 and every other entry is 0.

A **vector field** on $\mathbb{R}^n$ is a function that takes a point in $\mathbb{R}^n$ and spits out a vector in $\mathbb{R}^n$ whose initial point is the input.

Now let us consider matrices. An $m \times n$ **matrix** is a rectangular array of numbers, with m rows and n columns. We define matrix multiplication between an $m \times n$ and $n \times p$ matrix as follows: suppose that a_{ij} indicates the entry of a matrix A in the ith row and jth column, and similarly for b_{ij}. $C = AB$ is the $m \times p$ matrix with entries defined as

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj} \tag{2.59}$$

Theorem 12. *Matrix multiplication is associative; that is, whenever the matrices are defined in such a way that multiplication is applicable:*

$$(AB)C = A(BC) \tag{2.60}$$

Proof. One can verify that the i,jth entry of $(AB)C$ and $A(BC)$ depend only on the ith row of A and the jth column of C, so without loss of generality, we can assume A is a row matrix and C is a column matrix, so $(AB)C$ and $A(BC)$

are both numbers. Therefore, the proof reduces itself to a simple calculation:

$$(AB)C = \sum_{l=1}^{p} \left(\sum_{k=1}^{m} a_k b_{kl} \right) c_l \tag{2.61}$$

$$= \sum_{l=1}^{p} \sum_{k=1}^{m} a_k b_{kl} c_l = \sum_{k=1}^{m} a_k \left(\sum_{l=1}^{p} b_{kl} c_l \right) = A(BC) \tag{2.62}$$

□

Note that matrix multiplication is not commutative.

We define the **identity matrix** I_n as the $n \times n$ matrix with 1s along the main diagonal and 0s elsewhere.

Suppose A is a matrix. If

$$BA = I \tag{2.63}$$

then B is a left inverse of A. Likewise, if

$$AB = I \tag{2.64}$$

then B is a right inverse of A. A is **invertible** if it has both a left and right inverse.

Theorem 13. *If A is invertible, then its left inverse is equal to its right inverse, and this matrix is called the **inverse** of A, denoted A^{-1}.*

Proof. Suppose B is the left inverse of A and C is the right inverse of A. Then

$$B = BI = B(AC) = (BA)C = IC = C \tag{2.65}$$

Hence $B = C$. □

Theorem 14. *If A and B are invertible, then AB is invertible, where $(AB)^{-1} = B^{-1}A^{-1}$.*

Proof. We have

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I \tag{2.66}$$

and

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I \tag{2.67}$$

Therefore, $B^{-1}A^{-1}$ is both a left and right inverse of AB, and is therefore the inverse of AB. □

The **transpose** of an $m \times n$ matrix A, denoted A^T, is the $n \times m$ matrix with entries b_{ij} such that

$$b_{ij} = a_{ji} \tag{2.68}$$

Note that we have

$$(AB)^T = B^T A^T \tag{2.69}$$

There are other kinds of matrices that may be of importance. For instance, a **symmetric** matrix is a matrix A such that $A = A^T$. An anti-symmetric matrix is a matrix A such that $A = -A^T$. An **upper triangular** matrix is a matrix with nonzero entries only on or above the main diagonal; there is a similar definition for a **lower triangular** matrix. A **diagonal** matrix has nonzero entries only on the main diagonal.

We define the **determinant** of a 2 by 2 matrix to be

$$\det \begin{pmatrix} a_1 & b_1 \\ a_2 & b_2 \end{pmatrix} = a_1 b_2 - a_2 b_1 \tag{2.70}$$

It is a fact that the area of the parallelogram spanned by the vectors

$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}, \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \tag{2.71}$$

is the absolute value of $\det(\vec{\mathbf{a}}, \vec{\mathbf{b}})$. The determinant of a 3 by 3 matrix is defined as

$$\det \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix} = a_1 \det \begin{pmatrix} b_2 & c_2 \\ b_3 & c_3 \end{pmatrix} - a_2 \det \begin{pmatrix} b_1 & c_1 \\ b_3 & c_3 \end{pmatrix} + a_3 \det \begin{pmatrix} b_1 & c_1 \\ b_2 & c_2 \end{pmatrix} \tag{2.72}$$

One can create an analogous relation between the 3×3 determinant and the parallepiped as we have stated for the 2×2 determinant and the parallelogram.

Chapter 3

Basic Linear Algebra

Let us discuss mappings. A mapping associates each element of one set to an element of another. There must be a **domain**, which is the set of departure, a **range**, which is the set of arrival, and a rule which defines the mapping.

We say that a mapping is **onto**, or surjective, if every element in the range is associated with at least one element in the domain. Furthermore, a mapping is **one to one**, or injective, if every element in the range is associated with at most one element of the domain. Finally, a mapping is **bijective** if it is both injective and surjective; a bijective mapping is invertible.

Suppose that we have two mappings f and g. The composition of these two mappings $f \circ g$ is defined as

$$(f \circ g)(x) = f(g(x)) \tag{3.1}$$

Consider the following theorem below.

Theorem 15. *Composition is associative; that is,* $f \circ g \circ h = (f \circ g) \circ h = f \circ (g \circ h)$.

Proof. The proof is as follows:

$$((f \circ g) \circ h)(x) = (f \circ g)(h(x)) = f(g(h(x))) \tag{3.2}$$

and

$$(f \circ (g \circ h))(x) = f((g \circ h)(x)) = f(g(h(x))) \tag{3.3}$$

Hence $((f \circ g) \circ h)(x) = (f \circ (g \circ h))(x)$, and so composition is associative. □

We define a **linear transformation** $T : \mathbb{R}^n \to \mathbb{R}^m$ as a mapping such that for all scalars a and all $\vec{\mathbf{v}}, \vec{\mathbf{w}} \in \mathbb{R}^n$, we have that

$$T(\vec{\mathbf{v}} + \vec{\mathbf{w}}) = T(\vec{\mathbf{v}}) + T(\vec{\mathbf{w}}) \quad \text{and} \quad T(a\vec{\mathbf{v}}) = aT(\vec{\mathbf{v}}) \tag{3.4}$$

Interestingly enough, every linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ can be given by multiplication by an $m \times n$ matrix $[T]$, where the ith column of the matrix is $T(\vec{\mathbf{e}}_i)$. Furthermore, if we have two linear transformations $S : \mathbb{R}^n \to \mathbb{R}^m$ and $T : \mathbb{R}^m \to \mathbb{R}^l$ given by the matrices $[S]$ and $[T]$, then the matrix of the composition $T \circ S$ is given by the product of the matrices $[T][S]$. It necessarily follows from this fact and the fact that composition is associative that matrix multiplication as well is associative.

Chapter 4

Vector-Valued Functions

So a function of the form

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} \tag{4.1}$$

is called a **plane curve**, and a function of the form

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k} \tag{4.2}$$

is called a **space curve**. Both of these are called **vector-valued functions**. To take the limit of a vector-valued function (we will assume that it is a space curve, and the property can be easily extended to plane curves), we consider

$$\lim_{t\to a} \mathbf{r}(t) = \left(\lim_{t\to a} x(t)\right)\mathbf{i} + \left(\lim_{t\to a} y(t)\right)\mathbf{j} + \left(\lim_{t\to a} z(t)\right)\mathbf{k} \tag{4.3}$$

We say that a vector-valued function is continuous at a point where $t = a$ if

$$\lim_{t\to a} \mathbf{r}(t) = \mathbf{r}(a) \tag{4.4}$$

(Of course, both should exist at that point as well in order for the function to be continuous at that point.)

Now, we define the **derivative** of the vector-valued function $\mathbf{r}$ as

$$\mathbf{r}'(t) = \lim_{\Delta t \to 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \tag{4.5}$$

If $\mathbf{r}'(c)$ exists, then $\mathbf{r}$ is differentiable at the point $t = c$. Interestingly, we have the following theorem:

Theorem 16. *If we have $\boldsymbol{r}(t) = x(t)\boldsymbol{i} + y(t)\boldsymbol{j} + z(t)\boldsymbol{k} = \langle x(t), y(t), z(t)\rangle$, and $x(t), y(t), z(t)$ are all differentiable functions, then*

$$\boldsymbol{r}'(t) = \langle x'(t), y'(t), z'(t)\rangle \tag{4.6}$$

The proof is an easy exercise and is left to the reader.

$\mathbf{r}(t)$ is smooth on some open interval I if x', y', z' are continuous on I and $\mathbf{r}'(t) \neq 0$ for all $t \in I$.

The derivative of a vector-valued function has some interesting properties, which will be listed below:

Theorem 17. *Denoting $D_t(\boldsymbol{r}(t))$ as the derivative of $\boldsymbol{r}(t)$, we have*

1. $D_t(c\boldsymbol{r}(t)) = c\boldsymbol{r}'(t)$
2. $D_t(\boldsymbol{r}(t) \pm \boldsymbol{u}(t)) = \boldsymbol{r}'(t) \pm \boldsymbol{u}'(t)$
3. $D_t(f(t)\boldsymbol{r}(t)) = f(t)\boldsymbol{r}'(t) + f'(t)\boldsymbol{r}(t)$
4. $D_t(\boldsymbol{r}(t) \cdot \boldsymbol{u}(t)) = \boldsymbol{r}(t) \cdot \boldsymbol{u}'(t) + \boldsymbol{r}'(t) \cdot \boldsymbol{u}(t)$
5. $D_t(\boldsymbol{r}(t) \times \boldsymbol{u}(t)) = \boldsymbol{r}(t) \times \boldsymbol{u}'(t) + \boldsymbol{r}'(t) \times \boldsymbol{u}'(t)$
6. $D_t(\boldsymbol{r}(f(t))) = \boldsymbol{r}'(f(t))f'(t)$

7. *If* $\boldsymbol{r}(t) \cdot \boldsymbol{r}(t) = c$*, then* $\boldsymbol{r}(t) \cdot \boldsymbol{r}'(t) = 0$

Proof. We will prove only the last property, which is the least straightforward. We are given that

$$\mathbf{r}(t) \cdot \mathbf{r}(t) = c \tag{4.7}$$

It follows that

$$D_t(\mathbf{r}(t) \cdot \mathbf{r}(t)) = D_t(c) \tag{4.8}$$

Note that $D_t c = 0$, and by the product rule,

$$\mathbf{r}'(t) \cdot \mathbf{r}(t) + \mathbf{r}(t) \cdot \mathbf{r}'(t) = 0 \tag{4.9}$$

By the commutativity of the dot product,

$$2(\mathbf{r}'(t) \cdot \mathbf{r}(t)) = 0 \tag{4.10}$$

Dividing by 2 yields the desired expression. □

Just like with differentiation, we can define **integration** of vector-valued functions like so:

$$\int \mathbf{r}(t)dt = \left\langle \int x(t)dt, \int y(t)dt, \int z(t)dt \right\rangle \tag{4.11}$$

and

$$\int_a^b \mathbf{r}(t)dt = \left\langle \int_a^b x(t)dt, \int_a^b y(t)dt, \int_a^b z(t)dt \right\rangle \tag{4.12}$$

We now define three key terms in the study of parametric equations and curves. The **velocity** of a curve is equal to

$$\mathbf{v}(t) = \mathbf{r}'(t) \tag{4.13}$$

The **acceleration** of a curve is equal to

$$\mathbf{a}(t) = \mathbf{r}''(t) \tag{4.14}$$

Finally, the speed of a curve is equal to

$$||\mathbf{v}(t)|| = ||\mathbf{r}'(t)|| \tag{4.15}$$

Suppose that we have a projectile launched from an initial height h, with initial speed v_0, and an angle of elevation θ. The equation for its path is

$$\mathbf{r}(t) = \left\langle v_0 \cos\theta, h + (v_0 \sin\theta)t - \frac{1}{2}gt^2 \right\rangle \tag{4.16}$$

This comes from the fact that $\mathbf{a} = \langle 0, -g \rangle$, and integrating, considering the correct constants of integration and initial conditions as we go.

The **unit tangent** vector is defined to be

$$\mathfrak{t}(t) = \frac{\mathbf{r}'(t)}{||\mathbf{r}'(t)||}, \quad \mathbf{r}'(t) \neq \mathbf{0} \tag{4.17}$$

Meanwhile, the **unit normal vector** is defined to be

$$\mathfrak{n}(t) = \frac{\mathfrak{t}'(t)}{||\mathfrak{t}'(t)||} \tag{4.18}$$

We now consider a theorem regarding the acceleration vector:

Theorem 18. *The acceleration vector $\boldsymbol{a}(t)$ lies in the plane determined uniquely by $\mathfrak{t}(t)$ and $\mathfrak{n}(t)$.*

Proof. Let us represent $\mathbf{v} = ||\mathbf{v}||\mathfrak{t}$, since

$$\mathfrak{t} = \frac{\mathbf{v}}{||\mathbf{v}||} \tag{4.19}$$

Then, differentiating the former equation, we obtain

$$\mathbf{a} = D_t(||\mathbf{v}||)\mathfrak{t} + ||\mathbf{v}||\mathfrak{t}' \tag{4.20}$$

$$= D_t(||\mathbf{v}||)\mathfrak{t} + ||\mathbf{v}||||\mathfrak{t}'||\mathfrak{n} \tag{4.21}$$

which we obtain by considering $\mathfrak{n} = \mathfrak{t}/||\mathfrak{t}||$. Therefore, we have written the acceleration vector as a linear combination of the unit tangent and the unit normal, and hence, the acceleration vector lies in the plane determined by those two vectors. $\square$

The above theorem implies that there exist **tangential** and **normal components of acceleration**, defined reasonably like so; the tangential component of acceleration:

$$a_{\mathfrak{t}} = D_t(||\mathbf{v}||) = \frac{\mathbf{v} \cdot \mathbf{a}}{||\mathbf{v}||} \tag{4.22}$$

the normal, or centripetal component of acceleration:

$$a_{\mathfrak{n}} = ||\mathbf{v}||||\mathfrak{t}'|| = \frac{||\mathbf{v} \times \mathbf{a}||}{||\mathbf{v}||} \tag{4.23}$$

The **arc length** of a space curve on an interval $[a, b]$ is defined to be

$$s = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}dt = \int_a^b ||\mathbf{r}'(t)||dt \tag{4.24}$$

This naturally gives rise to an arc length function defined by

$$s(t) = \int_a^t ||\mathbf{r}'(u)||du = \int_a^t \sqrt{(x'(u))^2 + (y'(u))^2 + (z'(u))^2}du \tag{4.25}$$

Suppose we then have a curve defined by

$$\mathbf{r}(s) = \langle x(s), y(s), z(s) \rangle \tag{4.26}$$

where s is the arc length function. It follows that

$$||\mathbf{r}'(s)|| = 1 \tag{4.27}$$

We define curvature to be the rate of change of the unit tangent with respect to arc length:

$$\kappa = \left\| \frac{d\mathfrak{t}}{ds} \right\| = \|\mathfrak{t}'(s)\| \tag{4.28}$$

There are a couple formulas for curvature. The first is that

$$\kappa = \frac{\|\mathfrak{t}'(t)\|}{\|\mathbf{r}'(t)\|} = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3} \tag{4.29}$$

Furthermore, if we have a twice-differentiable function $y = f(x)$, the curvature at the point (x, y) is

$$\kappa = \frac{|y''|}{(1 + (y')^2)^{3/2}} \tag{4.30}$$

The circle which passes through a given point with the radius $1/\kappa$, where κ is the curvature at that point, is called the **circle of curvature**. The radius is called the **radius of curvature** and the center of the circle is called the **center of curvature**.

We can represent acceleration in terms of the arc length and curvature as well:

$$\mathbf{a}(t) = \frac{d^2 s}{dt^2}\mathfrak{t} + \kappa \left(\frac{ds}{dt}\right)^2 \mathfrak{n} \tag{4.31}$$

Chapter 5

Partial Differentiation

We begin by first defining a **function of two variables**, which is a mapping from a set D of ordered pairs (x, y) to some real number $f(x, y)$; we denote this mapping by f. The set D of ordered pairs is called the **domain**, and the set of values of $f(x, y)$ is called the **range**. We also have that, with this notation, x and y are considered to be **independent variables**, and if we have some $z = f(x, y)$, then z is the **dependent variable**. We define operations on functions like so:

$$(f \pm g)(x, y) = f(x, y) \pm g(x, y) \tag{5.1}$$

$$(fg)(x, y) = f(x, y)g(x, y) \tag{5.2}$$

$$\frac{f}{g}(x, y) = \frac{f(x, y)}{g(x, y)}, \quad g(x, y) \neq 0 \tag{5.3}$$

We can also define the **composite** function like so; given g is a single-variable function and h is a two variable function,

$$(g \circ h)(x, y) = g(h(x, y)) \tag{5.4}$$

A **polynomial** function in two variables is defined to be a function of the type

$$f(x,y) = \sum_{m,n} cx^m y^n, \quad m,n \geq 0 \tag{5.5}$$

Similarly, a **rational** function is simply the quotient of two polynomial functions.

We can define a **graph** for a function of two variables $f(x,y)$ with domain D as the set of all points (x,y,z) for which $z = f(x,y)$, where $(x,y) \in D$. We can also visualize a function by using a **scalar field**, where we assign a scalar $z = f(x,y)$ to a point (x,y) on a plane. **Level curves**, or **contour lines**, are curves in which $f(x,y)$ is constant. Level curves for a pressure function in weather maps are called **isobars**; for a temperature function, the level curves are called **isotherms**. For electric potential fields, level curves are called **equipotential lines**. **Topographic maps** also utilize level curves to show elevation from sea level.

The **Cobb-Douglas production function** is a function that gives the units produced when considering amounts of labor and capital; if x represents units of labor and y represents units of capital, we have

$$f(x,y) = Cx^a y^{1-a} \tag{5.6}$$

I had to do a project with the Cobb-Douglas production function when I was taking multivariable calculus.

Now suppose that $f(x,y,z)$ is a function of three variables; the graph of the equation $f(x,y,z) = c$ where c is a constant is called a **level surface**.

We define a **δ-neighborhood** around a point (x_0, y_0) to be the disk with radius $\delta > 0$. An **open disk** involves the following formula:

$$\left\{(x,y)|\sqrt{(x-x_0)^2+(y-y_0)^2}<\delta,\delta>0\right\} \tag{5.7}$$

A **closed disk** involves the following formula:

$$\left\{(x,y)|\sqrt{(x-x_0)^2+(y-y_0)^2}\leq\delta,\delta>0\right\} \tag{5.8}$$

More generally, the **open ball** with radius r around a point $\mathbf{x}\in\mathbb{R}^n$ is the subset

$$B_r(\mathbf{x})=\{\mathbf{y}\in\mathbb{R}^n|\|\mathbf{x}-\mathbf{y}\|<r\} \tag{5.9}$$

Suppose we have a subset of $\mathbb{R}^2$, called R. We say that a point (x,y) is an **interior point** in R if there exists some δ-neighborhood centered at (x,y) that is a subset of R. If every point in R is an interior point, then we define R to be an **open region**. In other words, a region, or set, U, is open if for every $\mathbf{x}\in U$ there exists some $r>0$ such that $B_r(\mathbf{x})\subset U$. A **boundary point** (x,y) is a point such that every open disk centered at (x,y) contains points inside R as well as points outside R at the same time. If the set R contains every single one of its boundary points, then we define R to be **closed**. A closed set can also be thought of as a set whose complement is open. Did you know that some sets can be both open and closed? We call them **clopen**. Two examples are the empty set $\emptyset$ and $\mathbb{R}^2$ (at least, in $\mathbb{R}^2$).

With this knowledge, we can now rigorously define the limit of a function of two variables. We have that

$$\lim_{(x,y)\to(x_0,y_0)}f(x,y)=L \tag{5.10}$$

if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta \implies |f(x, y) - L| < \varepsilon \tag{5.11}$$

Note how this is similar to the one-variable epsilon-delta definition of a limit. To verify that a limit exists by the epsilon-delta method, it is usually best to work backwards. That is, suppose $|f(x, y) - L| < \varepsilon$ and work backwards until you obtain an expression $\sqrt{(x - x_0)^2 + (y - y_0)^2}$, which should end up being less than epsilon multiplied by something. Set $\delta =$ whatever you got, and the proof follows easily from there.

Now let's talk about continuity. A function of two variables is **continuous** at a point (x_0, y_0) in an open region R if

$$\lim_{(x,y)\to(x_0,y_0)} f(x, y) = f(x_0, y_0) \tag{5.12}$$

and if it is continuous at every point in R, then the function is continuous on R. Continuity exhibits some nice properties:

Theorem 19. *If we have that f and g are continuous at (x_0, y_0) and $k \in \mathbb{R}$, then these functions are continuous at (x_0, y_0):*

1. kf
2. $f \pm g$
3. fg
4. f/g, *if* $g(x_0, y_0) \neq 0$

The proof will not be provided, for the sake of brevity.

Furthermore, we can say that if g is continuous at $h(x_0, y_0)$ and h is continuous at (x_0, y_0), it follows that $(g \circ h)(x, y)$ is continuous at (x_0, y_0).

Everything stated before can be extended to three (or higher) dimensions. For instance, instead of considering two-dimensional disks, we may consider three-dimensional spheres to be the δ-neighborhood of a point:

$$(x-x_0)^2+(y-y_0)^2+(z-z_0)^2<\delta^2 \tag{5.13}$$

We say that a subset $C \subset \mathbb{R}^n$ is **compact** if it is closed and bounded. Topologically, the technically correct definition is that a set is compact if any open covering of the set admits a finite subcovering, but you don't need to understand that. The **Heine-Borel theorem** makes it very clear that in $\mathbb{R}^n$, our previous definition suffices.

Now we come to the definition of a **partial derivative**. For functions of two variables, we have that

$$f_x(x,y)=\lim_{\Delta x\to 0}\frac{f(x+\Delta x,y)-f(x,y)}{\Delta x} \tag{5.14}$$

and

$$f_y(x,y)=\lim_{\Delta y\to 0}\frac{f(x,y+\Delta y)-f(x,y)}{\Delta y} \tag{5.15}$$

Let us discuss notation. We can use the following notation for the partial derivatives of the function $z=f(x,y)$:

$$f_x(x,y)=\frac{\partial}{\partial x}f(x,y)=z_x=\frac{\partial z}{\partial x} \tag{5.16}$$

$$f_y(x,y)=\frac{\partial}{\partial y}f(x,y)=z_y=\frac{\partial z}{\partial y} \tag{5.17}$$

To evaluate partial derivatives at a point (a,b), we say

$$f_x(a,b)=\left.\frac{\partial z}{\partial x}\right|_{(a,b)} \quad \text{and} \quad f_y(a,b)=\left.\frac{\partial z}{\partial y}\right|_{(a,b)} \tag{5.18}$$

Partial derivatives essentially give the slopes of the surface defined by $z = f(x, y)$ in the x and y directions, respectively.

Now let's talk about higher-order partial derivatives. To differentiate twice with respect to x, we write

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} = f_{xx} \tag{5.19}$$

To differentiate twice with respect to y, we write

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2} = f_{yy} \tag{5.20}$$

To differentiate with respect to x first and then differentiate with respect to y, we write

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x} = f_{xy} \tag{5.21}$$

Be careful with the notation! Differentiating with respect to y first and then with respect to x is similar. We call the latter two **mixed partial derivatives**. Interestingly, we have that:

Theorem 20. *If f_{xy} and f_{yx} are continuous on some open set R, then, for all $(x, y) \in R$, we have that*

$$f_{xy}(x, y) = f_{yx}(x, y) \tag{5.22}$$

We define an **increment** of x as Δx, and we define it for y similarly. If we have $z = f(x, y)$, the increment of z Δz is defined to be

$$\Delta z = f(x + \Delta x, y + \Delta y) - f(x, y) \tag{5.23}$$

So, we define **differentials** of x and y to be $dx = \Delta x$ and $dy = \Delta y$, whereas the **total differential** of $z = f(x,y)$ is

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy \tag{5.24}$$

This should ring a bell from single-variable calculus, except it is extended to higher dimensions. We can extend the definition of a two-variable differential to three variables and beyond.

Now we define **differentiability**. We say that $z = f(x,y)$ is differentiable at a point (x_0, y_0) if we can write Δz as

$$\Delta z = f_x(x_0,y_0)\Delta x + f_y(x_0,y_0)\Delta y + \varepsilon_1 \Delta x + \varepsilon_2 \Delta y \tag{5.25}$$

for some $\varepsilon_1, \varepsilon_2$ which both approach 0 as $(\Delta x, \Delta y) \to (0,0)$. The idea behind this definition is that if a function is differentiable, we can approximate dz by Δz, and as the increments are smaller and smaller, the approximation gets better and better. It just so happens that:

Theorem 21. *If f_x and f_y are continuous on some open set R, then f is differentiable on the open set.*

We can approximate Δz, and if we do so via the differential dz, we call that a **linear approximation**.

We consider another theorem, which should be familiar from single-variable calculus. The result still holds true here.

Theorem 22. *If f is differentiable at (x_0, y_0), then f is continuous at (x_0, y_0).*

Let's talk about the **chain rule**. We consider special cases. Suppose $w = f(x,y)$, and $x = g(t)$ and $y = h(t)$. Then

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} \tag{5.26}$$

Now suppose $w = f(x, y)$, $x = g(s, t)$, and $y = h(s, t)$. Then

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial s} \tag{5.27}$$

and

$$\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial t} \tag{5.28}$$

The punchline is that in the general case,

$$\frac{\partial w}{\partial t_p} = \sum_{i=1}^{n} \frac{\partial w}{\partial x_i}\frac{\partial x_i}{\partial t_p} \tag{5.29}$$

You might be wondering if there is a connection between partial differentiation and implicit differentiation, and as it turns out, there actually is! Suppose we have some $F(x, y) = 0$. Then

$$\frac{dy}{dx} = -\frac{F_x(x, y)}{F_y(x, y)} \tag{5.30}$$

and if we have some $F(x, y, z) = 0$, then

$$\frac{\partial z}{\partial x} = -\frac{F_x(x, y, z)}{F_z(x, y, z)}, \quad \frac{\partial z}{\partial y} = -\frac{F_y(x, y, z)}{F_z(x, y, z)} \tag{5.31}$$

Chapter 6

Applications of Partial Differentiation

Let us begin by discussing the **directional derivative** of a function $f(x, y)$ in the direction of the unit vector $\mathbf{u} = \langle \cos\theta, \sin\theta \rangle$. We have

$$D_{\mathbf{u}} f(x, y) = \lim_{t \to 0} \frac{f(x + t\cos\theta, y + t\sin\theta) - f(x, y)}{t} \tag{6.1}$$

The directional derivative can be calculated in a rather simple manner.

Theorem 23. *Suppose $f(x, y)$ is differentiable. Then the directional derivative of f in the direction of the vector $\boldsymbol{u} = \langle \cos\theta, \sin\theta \rangle$ is*

$$D_{\boldsymbol{u}} f(x, y) = f_x(x, y)\cos\theta + f_y(x, y)\sin\theta \tag{6.2}$$

Proof. Define $x(t) = x_0 + t\cos\theta$ and $y(t) = y_0 + t\sin\theta$. If $g(t) = f(x, y)$, then differentiating by the chain rule gives

$$g'(t) = f_x(x, y)x'(t) + f_y(x, y)y'(t) = f_x(x, y)\cos\theta + f_y(x, y)\sin\theta \tag{6.3}$$

Setting $t = 0$ gives

$$g'(0) = f_x(x_0, y_0)\cos\theta + f_y(x_0, y_0)\sin\theta \tag{6.4}$$

Meanwhile, the limit definition of the derivative for $g(0)$ yields

$$g'(0) = \lim_{t\to 0} \frac{g(t) - g(0)}{t} \tag{6.5}$$

$$= \lim_{t\to 0} \frac{f(x_0 + t\cos\theta, y_0 + t\sin\theta) - f(x_0, y_0)}{t} \tag{6.6}$$

By equating the various expressions equal to $g'(0)$, we find that

$$D_{\mathbf{u}} f(x_0, y_0) = f_x(x_0, y_0)\cos\theta + f_y(x_0, y_0)\sin\theta \tag{6.7}$$

□

Now we define the **gradient** of a function of two variables, $f(x, y)$, as

$$\nabla f(x, y) = \text{grad} f(x, y) = \langle f_x(x, y), f_y(x, y)\rangle \tag{6.8}$$

Using the gradient, we can define the directional derivative alternatively, like so:

Theorem 24. *If $f(x, y)$ is differentiable, then*

$$D_{\boldsymbol{u}} f(x, y) = \nabla f(x, y) \cdot \boldsymbol{u} \tag{6.9}$$

Proof. We know that

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y)\rangle \tag{6.10}$$

And

$$\mathbf{u} = \langle \cos\theta, \sin\theta \rangle \tag{6.11}$$

as defined before. The dot product of these two vectors must be

$$\nabla f(x,y) \cdot \mathbf{u} = f_x(x,y)\cos\theta + f_y(x,y)\sin\theta = D_{\mathbf{u}} f(x,y) \tag{6.12}$$

□

The gradient enjoys several properties, which we are able to make use of thoughtfully:

1. If we have that $\nabla f(x,y) = \mathbf{0}$, then the directional derivative must vanish for all vectors.

2. $\nabla f(x,y)$ points in the direction of maximum increase of the function f. It follows that the maximum value taken by $D_{\mathbf{u}} f(x,y)$ is $\|\nabla f(x,y)\|$.

3. Similarly, $-\nabla f(x,y)$ points in the direction of minimum increase of the function f. Furthermore, the minimum value of $D_{\mathbf{u}} f(x,y)$ is simply $-\|\nabla f(x,y)\|$.

We also have the following:

Theorem 25. *If we have that $\nabla f(x_0,y_0) \neq \mathbf{0}$, then $\nabla f(x_0,y_0)$ is normal to the level curve through (x_0,y_0).*

Proof. Suppose that we have a vector $\mathbf{u}$ pointing in the direction of the level curve. The directional derivative in the direction of $\mathbf{u}$ must be 0. Therefore,

$$D_{\mathbf{u}} f(x,y) = \nabla f(x,y) \cdot \mathbf{u} = 0 \tag{6.13}$$

which, as shown above, implies that the vector $\mathbf{u}$ is normal to that of the gradient, and hence the gradient is normal to the level curve. □

We can define the directional derivative and gradient for three variables as well, like so:

$$D_{\mathbf{u}}f(x,y,z) = af_x(x,y,z) + bf_y(x,y,z) + cf_z(x,y,z) \tag{6.14}$$

where $\mathbf{u} = \langle a, b, c\rangle$ is a *unit* vector. Furthermore,

$$\nabla f(x,y,z) = \langle f_x(x,y,z), f_y(x,y,z), f_z(x,y,z)\rangle \tag{6.15}$$

The properties above in this chapter extend to that of three dimensions. It is simple to verify this claim.

We now proceed to define the **tangent plane** to a surface S given by $F(x,y,z) = 0$ where $\nabla F(x,y,z) \neq \mathbf{0}$: the plane uniquely defined by being normal to $\nabla F(x_0, y_0, z_0)$ is called the tangent plane to S at (x_0, y_0, z_0). The **normal line** to S at (x_0, y_0, z_0) is the line in the direction of $\nabla F(x_0, y_0, z_0)$.

The tangent plane has an equation that we can use:

Theorem 26. *The tangent plane to the surface given by* $F(x,y,z) = 0$ *at the point* (x_0, y_0, z_0) *is given by the equation*

$$F_x(x_0,y_0,z_0)(x-x_0) + F_y(x_0,y_0,z_0)(y-y_0) + F_z(x_0,y_0,z_0)(z-z_0) = 0 \tag{6.16}$$

Recall the standard unit vector $\mathbf{k} = \langle 0, 0, 1\rangle$. We can find the **angle of inclination** of a plane (the angle between the plane and the xy-plane) by

$$\cos\theta = \frac{|\mathbf{n}\cdot\mathbf{k}|}{\|\mathbf{n}\|} \tag{6.17}$$

It should be clear at this point that if $\nabla F(x_0, y_0, z_0) \neq \mathbf{0}$, then $\nabla F(x_0, y_0, z_0)$ is normal to the level surface through (x_0, y_0, z_0), in a similar manner as to how the gradient of a function in two variables is (under certain conditions) normal to the level curve through the point as well.

A **bounded** region must be a subset of a closed disk in the plane; that is, there must exist some disk out there that all elements in the region are in the disk as well. Suppose that we have a function f on a closed and bounded region R. The values $f(a,b)$ and $f(c,d)$ such that

$$f(a,b) \leq f(x,y) \leq f(c,d) \tag{6.18}$$

for all $(x,y) \in R$ are called the **minimum** and **maximum** of the function in R, respectively. We can also define relative minima and maxima; a **relative minimum** $f(x_0, y_0)$ at (x_0, y_0) is a value such that

$$f(x,y) \geq f(x_0, y_0) \tag{6.19}$$

for all (x,y) in an *open* disk containing (x_0, y_0). The **relative maximum** $f(x_0, y_0)$ at (x_0, y_0) is defined similarly, where

$$f(x,y) \leq f(x_0, y_0) \tag{6.20}$$

We define a **critical point** as any point (x_0, y_0) where one of the following must be true:

1. $f_x(x_0, y_0) = 0$ AND $f_y(x_0, y_0) = 0$
2. $f_x(x_0, y_0)$ OR $f_y(x_0, y_0)$ does not exist

Relative extrema on an open region R can only be found on critical points of f.

Now, if you have a region that contains boundary points, you will have to check both critical points and boundary points as well to find the extrema on the region.

A **saddle point** is a critical point which has neither a relative maximum nor a relative minimum. To determine what kind of extremum we are dealing with, the **second partials test** comes in handy.

1. Suppose $f_x(a,b) = 0$ and $f_y(a,b) = 0$.
2. Compute $d = f_{xx}(a,b)f_{yy}(a,b) - [f_{xy}(a,b)]^2$.
3. If $d > 0$ and $f_{xx}(a,b) > 0$, then f has a **relative minimum** at (a,b).
4. If $d > 0$ and $f_{xx}(a,b) < 0$, then f has a **relative maximum** at (a,b).
5. If $d < 0$, then $(a,b,f(a,b))$ is a **saddle point**.
6. If $d = 0$, the test is inconclusive.

You can think of

$$d = \begin{vmatrix} f_{xx}(a,b) & f_{xy}(a,b) \\ f_{yx}(a,b) & f_{yy}(a,b) \end{vmatrix} \tag{6.21}$$

Let's talk about the **method of least squares**. To find a line of best fit, we can think about the sum of the squared errors

$$S = \sum_{i=1}^{n} [f(x_i) - y_i]^2 \tag{6.22}$$

Our goal is to optimize this by finding a minimum. As it turns out, there does exist a general solution to this problem:

Theorem 27. *The* ***least squares regression line*** *for a set of points $\{(x_i, y_i)\}$ is given by the function $f(x) = ax + b$, where*

$$a = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2} \tag{6.23}$$

$$b = \frac{1}{n}\left(\sum_{i=1}^{n} y_i - a\sum_{i=1}^{n} x_i\right) \tag{6.24}$$

Proof. Here is a sketch of a proof. Calculate partial derivatives of S and set them equal to 0. Use methods of linear algebra to find a and b. Using the Second Partials Test, you can verify that these values of a and b yield a minimum. □

When it comes to optimization, sometimes there are constraints. In order to deal with these constraints, **Lagrange's theorem** and Lagrange multipliers will come in handy.

Theorem 28. *Suppose we wish to optimize a function $f(x, y)$ on the smooth constraint curve $g(x, y) = c$. If $\nabla g(x_0, y_0) \neq \mathbf{0}$, then there must exist some $\lambda \in \mathbb{R}$ such that*

$$\nabla f(x, y) = \lambda \nabla g(x, y) \tag{6.25}$$

Now if f and g have continuous first partial derivatives, to find a minimum or maximum of f on the curve $g(x, y) = c$, we solve the equations $\nabla f(x, y) = \lambda \nabla g(x, y)$ and $g(x, y) = c$. This should yield solution points (x, y). Evaluate f at all of these points, and you will find that the largest is the maximum, and the smallest is the minimum.

Chapter 7

Multiple Integration

To begin, we consider that we can integrate partial derivatives like so:

$$\int_{h_1(y)}^{h_2(y)} \frac{\partial f}{\partial x} dx = f(x,y)\Big|_{h_1(y)}^{h_2(y)} = f(h_2(y),y) - f(h_1(y),y) \tag{7.1}$$

or

$$\int_{g_1(x)}^{g_2(x)} \frac{\partial f}{\partial y} dy = f(x,y)\Big|_{g_1(x)}^{g_2(x)} = f(x,g_2(x)) - f(x,g_1(x)) \tag{7.2}$$

We define an **iterated integral** to be an expression with multiple integrals in succession, such as

$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)dydx \quad \text{or} \quad \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)dxdy \tag{7.3}$$

Note that the inside limits of integration must only be variable to outer variables of integration, whereas the outside limits of integration must be constant with respect to all variables of integration. For instance, we must have

$$\int_a^b \int_{f_1(z)}^{f_2(z)} \int_{g_1(y,z)}^{g_2(y,z)} h(x,y,z)dxdydz \tag{7.4}$$

The intervals found on the integrals determine the region of integration of the iterated integral. If we have a region bounded by the vertical lines $x = a$ and $x = b$ for some $a, b \in \mathbb{R}$, then the region is called **vertically simple**. A region bounded by the horizontal lines $y = a$ and $y = b$ for some $a, b \in \mathbb{R}$ is called **horizontally simple**.

To find the area of a region in the plane, we consider the following:

1. If the region is defined by $a \leq x \leq b$ and $g_1(x) \leq y \leq g_2(x)$, where g_1, g_2 are continuous on $[a, b]$, then the area can be calculated by

$$A = \int_a^b \int_{g_1(x)}^{g_2(x)} dydx \tag{7.5}$$

2. If the region is defined by $c \leq y \leq d$ and $h_1(y) \leq x \leq h_2(y)$, where h_1, h_2 are continuous on $[c, d]$, then the area can be calculated by

$$A = \int_c^d \int_{h_1(y)}^{h_2(y)} dxdy \tag{7.6}$$

To integrate a function of two variables over a region in the plane, we must employ **double integrals**. We define an **inner partition** Δ to be comprised of all rectangles in a rectangular grid over the region of integration, whereas the **norm** $\|\Delta\|$ of the partition is the length of the longest *diagonal* of the rectangles. In each rectangle, we choose a point (x_i, y_i) in the rectangle and form a rectangular prism for each rectangle. The area of the ith rectangle is defined to be ΔA_i, so the volume of the ith prism is $f(x_i, y_i)\Delta A_i$. We can see, in a similar manner to single-variable integration, that approximating the volume of the solid region gives

$$\sum_i f(x_i, y_i)\Delta A_i \tag{7.7}$$

This formula naturally lends itself to the definition of the double integral, which is similar in nature to the single-variable integral:

$$\iint_R f(x,y)dA = \lim_{\|\Delta\|\to 0} \sum_{i=1}^{n} f(x_i,y_i)\Delta A_i \tag{7.8}$$

If the right-hand limit exists, then we say that f is **integrable** over the region R. So, it is clear that the volume of the region that lies above R on the plane and below the graph of f is defined as

$$V = \iint_R f(x,y)dA \tag{7.9}$$

Double integrals have certain properties that we may employ:

Theorem 29. *1.* $\iint_R cf(x,y)dA = c\iint_R f(x,y)dA$

2. $\iint_R [f(x,y) \pm g(x,y)]dA = \iint_R f(x,y)dA \pm \iint_R g(x,y)dA$

3. $\iint_R f(x,y)dA \geq 0$ *if* $f(x,y) \geq 0$

4. $\iint_R f(x,y)dA \geq \iint_R g(x,y)dA$ *if* $f(x,y) \geq g(x,y)$

5. $\iint_R f(x,y)dA = \iint_{R_1} f(x,y)dA + \iint_{R_2} f(x,y)dA$ *if* R *is the union of two nonoverlapping regions* R_1 *and* R_2

We now come to **Fubini's theorem**, which allows us to evaluate double integrals as iterated integrals:

Theorem 30. *Suppose* f *is continuous on* R*, then*

1. *If* R *is defined by* $a \leq x \leq b$ *and* $g_1(x) \leq y \leq g_2(x)$*, and* g_1, g_2 *are continuous on* $[a,b]$ *then*

$$\iint_R f(x,y)dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)dydx \tag{7.10}$$

2. If R is defined by $c \leq y \leq d$ and $h_1(y) \leq x \leq h_2(y)$, and h_1, h_2 are continuous on $[c, d]$, then

$$\iint_R f(x,y)dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)dxdy \tag{7.11}$$

To convert between rectangular coordinates and polar coordinates, we have that if R is bounded by $0 \leq g_1(\theta) \leq r \leq g_2(\theta)$ and $\alpha \leq \theta \leq \beta$, where $0 \leq (\beta - \alpha) \leq 2\pi$, then

$$\iint_R f(x,y)dA = \int_\alpha^\beta \int_{g_1(\theta)}^{g_2(\theta)} f(r\cos\theta, r\sin\theta) r dr d\theta \tag{7.12}$$

A **triple integral** can be constructed in a very similar way to the double integral. We consider an **inner partition** of boxes lying entirely within a solid region Q. The volume of the ith box is

$$\Delta V_i = \Delta x_i \Delta y_i \Delta z_i \tag{7.13}$$

and the **norm** $\|\Delta\|$ is defined to be the length of the longest diagonal of the n boxes in the partition. Consider the Riemann sum

$$\sum_{i=1}^{n} f(x_i, y_i, z_i)\Delta V_i \tag{7.14}$$

as $\|\Delta\| \to 0$, we obtain the definition of the triple integral of f over Q:

$$\iiint_Q f(x,y,z)dV = \lim_{\|\Delta\| \to 0} \sum_{i=1}^{n} f(x_i, y_i, z_i)\Delta V_i \tag{7.15}$$

This naturally defines the **volume** of Q to be

$$\text{Volume} = \iiint_Q dV \tag{7.16}$$

The triple integral has some properties of note:

$$\iiint_Q cf(x,y,z)dV = c\iiint_Q f(x,y,z)dV \tag{7.17}$$

$$\iiint_Q [f(x,y,z) \pm g(x,y,z)]dV = \iiint_Q f(x,y,z)dV \pm \iiint_Q g(x,y,z)dV \tag{7.18}$$

$$\iiint_Q f(x,y,z)dV = \iiint_{Q_1} f(x,y,z)dV + \iiint_{Q_2} f(x,y,z)dV \tag{7.19}$$

where the union of two disjoint sets Q_1 and Q_2 is Q. Suppose that Q is defined by

$$a \le x \le b, \quad h_1(x) \le y \le h_2(x), \quad g_1(x,y) \le z \le g_2(x,y) \tag{7.20}$$

Then

$$\iiint_Q f(x,y,z)dV = \int_a^b \int_{h_1(x)}^{h_2(x)} \int_{g_1(x,y)}^{g_2(x,y)} f(x,y,z)dzdydx \tag{7.21}$$

The variables can be put in other orders, depending on how the region is bounded.

We can also find the triple integral in cylindrical form to be

$$\iiint_Q f(x,y,z)dV = \int_{\theta_1}^{\theta_2} \int_{g_1(\theta)}^{g_2(\theta)} \int_{h_1(r\cos\theta, r\sin\theta)}^{h_2(r\cos\theta, r\sin\theta)} f(r\cos\theta, r\sin\theta, z)rdzdrd\theta \tag{7.22}$$

if Q is bounded by $\theta_1 \le \theta \le \theta_2$, $g_1(\theta) \le r \le g_2(\theta)$, $h_1(r\cos\theta, r\sin\theta) \le z \le h_2(r\cos\theta, r\sin\theta)$. In spherical coordinates, we have

$$\iiint_Q f(x,y,z)dV$$

$$= \int_{\theta_1}^{\theta_2} \int_{\phi_1}^{\phi_2} \int_{\rho_1}^{\rho_2} f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\rho^2 \sin\phi d\rho d\phi d\theta \tag{7.23}$$

Note that the order of variables can vary depending on how the region is bounded.

Chapter 8

Applications of Multiple Integration

We can use integration to find the **mass** of an object, given a **mass density**. We have that, if ρ is a density function on a plane region R (which we call a **lamina**), then we calculate the mass by performing the integral

$$m = \iint_R \rho(x, y) dA \tag{8.1}$$

The **first moments** of mass with respect to the x and y axes are calculated by

$$M_x = \iint_R y\rho(x, y) dA \quad M_y = \iint_R x\rho(x, y) dA \tag{8.2}$$

Furthermore, the **center of mass** is the point

$$(\overline{x}, \overline{y}) = \left(\frac{M_y}{m}, \frac{M_x}{m}\right) \tag{8.3}$$

If R is a simple plane region, then the above point is known as the **centroid** of the region. The **second moments**, or **moments of inertia**, are defined about the x and y axes as follows:

$$I_x = \iint_R y^2 \rho(x,y) dA \quad I_y = \iint_R x^2 \rho(x,y) dA \tag{8.4}$$

which is a measure of the tendency of matter to resist a change in rotational motion. Note that y is the distance to the x-axis, and x is the distance to the y-axis. The **polar moment of inertia** $I_0 = I_x + I_y$. Finally, the **radius of gyration** $\overline{\overline{r}}$ of some mass m with a moment of inertia I is

$$\overline{\overline{r}} = \sqrt{\frac{I}{m}} \tag{8.5}$$

To calculate the **upper surface area** of a solid, consider dividing the surface into many partitions and approximating by parallelograms where the sides are given by the vectors $\mathbf{u} = \langle \Delta x_i, 0, f_x(x_i, y_i)\Delta x_i \rangle$ and $\mathbf{v} = \langle 0, \Delta y_i, f_y(x_i, y_i)\Delta y_i \rangle$. This makes more sense if you draw it out, so try drawing it out. Recall that the area of the parallelogram defined by those two vectors is $\|\mathbf{u} \times \mathbf{v}\|$, which becomes

$$\|\mathbf{u} \times \mathbf{v}\| = \sqrt{[f_x(x_i, y_i)]^2 + [f_y(x_i, y_i)]^2 + 1}\Delta A_i \tag{8.6}$$

where $\Delta A_i = \Delta x_i \Delta y_i$. Therefore, the approximate surface area of the surface is

$$\sum_{i=1}^{n} \sqrt{1 + [f_x(x_i, y_i)]^2 + [f_y(x_i, y_i)]^2}\Delta A_i \tag{8.7}$$

which implies that the actual surface area can be computed by the formula

$$\text{Surface area} = \iint_R \sqrt{1 + [f_x(x, y)]^2 + [f_y(x, y)]^2} dA \tag{8.8}$$

Suppose that there exists a density function ρ on a solid region Q. Then the **mass** is defined by the triple integral

$$m = \iiint_Q \rho(x,y,z)dV \tag{8.9}$$

The **first moments** are defined by

$$M_{yz} = \iiint_Q x\rho(x,y,z)dV \tag{8.10}$$

$$M_{xz} = \iiint_Q y\rho(x,y,z)dV \tag{8.11}$$

$$M_{xy} = \iiint_Q z\rho(x,y,z)dV \tag{8.12}$$

so the **center of mass** is

$$(\overline{x},\overline{y},\overline{z}) = \left(\frac{M_{yz}}{m}, \frac{M_{xz}}{m}, \frac{M_{xy}}{m}\right) \tag{8.13}$$

The **second moments**, or **moments of inertia**, are defined as

$$I_x = \iiint_Q (y^2+z^2)\rho(x,y,z)dV \tag{8.14}$$

$$I_y = \iiint_Q (x^2+z^2)\rho(x,y,z)dV \tag{8.15}$$

$$I_z = \iiint_Q (x^2+y^2)\rho(x,y,z)dV \tag{8.16}$$

Note that we also have

$$I_{xy} = \iiint_Q z^2\rho(x,y,z)dV \tag{8.17}$$

$$I_{xz} = \iiint_Q y^2\rho(x,y,z)dV \tag{8.18}$$

$$I_{xy} = \iiint_Q x^2 \rho(x, y, z) dV \tag{8.19}$$

and

$$I_x = I_{xz} + I_{xy}, \quad I_y = I_{xy} + I_{yz}, \quad I_z = I_{xz} + I_{yz} \tag{8.20}$$

We now define the **Jacobian** of $x = g(u, v)$ and $y = h(u, v)$, denoted by $\partial(x, y)/\partial(u, v)$, which is

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u} \tag{8.21}$$

So, for double integrals, we have

$$\iint_R f(x, y) dx dy = \iint_S f(g(u, v), h(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv \tag{8.22}$$

where S is the set of all (u, v) such that $(g(u, v), h(u, v)) \in R$. The proof will be omitted.

Chapter 9

Vector Calculus

A **vector field** over some region R is a function

$$\mathbf{F}(x, y, z) = \langle M(x, y, z), N(x, y, z), P(x, y, z)\rangle \tag{9.1}$$

where M, N, P are all functions of three variables. A vector field over a two-dimensional region is defined analogously.

Suppose that we have a position vector $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$. An **inverse square field** is a vector field of the form

$$\mathbf{F}(x, y, z) = \frac{k}{\|\mathbf{r}\|^2}\mathbf{u} \tag{9.2}$$

where $k \in \mathbb{R}$ and $\mathbf{u} = \mathbf{r}/\|\mathbf{r}\|$ which is the unit vector in the direction of $\mathbf{r}$.

We say that a vector field is **conservative** if there exists some differentiable function f such that $\mathbf{F} = \nabla f$. There is a test that allows you to figure out easily if a two-dimensional vector field is conservative or not; if

$$\frac{\partial N}{\partial x} = \frac{\partial M}{\partial y} \tag{9.3}$$

for a vector field $\mathbf{F} = \langle M, N \rangle$, then $\mathbf{F}$ is conservative.

We define the **curl** of a three-dimensional vector field like so:

$$\nabla \times \mathbf{F}(x,y,z) = \left\langle \frac{\partial P}{\partial y} - \frac{\partial N}{\partial z}, \frac{\partial M}{\partial z} - \frac{\partial P}{\partial x}, \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right\rangle \tag{9.4}$$

One can think of the curl intuitively as the following "determinant":

$$\nabla \times \mathbf{F}(x,y,z) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M & N & P \end{vmatrix} \tag{9.5}$$

If $\nabla \times \mathbf{F} = \mathbf{0}$, then $\mathbf{F}$ is **irrotational.** Furthermore, we have that for a three-dimensional vector field $\mathbf{F}$, it is true that $\mathbf{F}$ is conservative if and only if

$$\nabla \times \mathbf{F}(x,y,z) = \mathbf{0} \tag{9.6}$$

The **divergence** of a two-dimensional vector field is

$$\nabla \cdot \mathbf{F}(x,y) = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \tag{9.7}$$

The divergence of a three-dimensional vector field is

$$\nabla \cdot \mathbf{F}(x,y,z) = \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z} \tag{9.8}$$

If $\nabla \cdot \mathbf{F} = 0$, then $\mathbf{F}$ is called **divergence free**. Furthermore, we have that

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0 \tag{9.9}$$

no matter what. This fact follows from a simple calculation as well as the equality of mixed partial derivatives.

We have that some curve C is considered to be **piecewise smooth** if the inverval $[a, b]$ that is the domain of C can be divided up into a finite number of subintervals, on which C must be smooth.

The **line integral** of a plane curve is defined as

$$\int_C f(x,y)ds = \lim_{\|\Delta\|\to 0} \sum_{i=1}^{n} f(x_i, y_i)\Delta s_i \tag{9.10}$$

and for space curves, we have

$$\int_C f(x,y,z)ds = \lim_{\|\Delta\|\to 0} \sum_{i=1}^{n} f(x_i, y_i, z_i)\Delta s_i \tag{9.11}$$

To evaluate line integrals, we have that

$$\int_C f(x,y)ds = \int_a^b f(x(t), y(t))\sqrt{[x'(t)]^2 + [y'(t)]^2}dt \tag{9.12}$$

and

$$\int_C f(x,y,z)ds = \int_a^b f(x(t), y(t), z(t))\sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2} \tag{9.13}$$

If $f(x, y, z) = 1$, we get the arc length of the curve as the line integral.

Suppose that we have a continuous vector field $\mathbf{F}$ on a smooth curve C given by the expression $\mathbf{r}(t)$. We have that

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(x(t), y(t), z(t)) \cdot \mathbf{r}'(t)dt \tag{9.14}$$

Line integrals can sometimes be written in **differential form**:

$$\int_C M dx + N dy + P dz = \int_a^b \left(M\frac{dx}{dt} + N\frac{dy}{dt} + P\frac{dz}{dt} \right) dt \tag{9.15}$$

The **fundamental theorem of line integrals** allows us to compute line integrals easily:

Theorem 31. *Suppose we have a piecewise smooth curve C given by $\boldsymbol{r}(t) = \langle x(t), y(t)\rangle$, where $a \leq t \leq b$. If we have a conservative vector field $\boldsymbol{F}(x, y) = \langle M, N\rangle$, where M and N are continuous on a region R where $C \subset R$, then we must have that*

$$\int_C \boldsymbol{F}\cdot d\boldsymbol{r} = \int_C \nabla f \cdot d\boldsymbol{r} = f(x(b), y(b)) - f(x(a), y(a)) \tag{9.16}$$

where f is the potential function of $\boldsymbol{F}$.

Proof. Suppose that C is smooth. Then $\mathbf{F}(x, y) = \langle f_x(x, y), f_y(x, y)\rangle$, so

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \mathbf{F}\cdot \frac{d\mathbf{r}}{dt}dt = \int_a^b \left[f_x(x, y)\frac{dx}{dt} + f_y(x, y)\frac{dy}{dt}\right] dt \tag{9.17}$$

Utilizng the Chain Rule, we obtain that

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \frac{d}{dt}f(x(t), y(t))dt \tag{9.18}$$

and by the Fundamental Theorem of Calculus, we have

$$= f(x(b), y(b)) - f(x(a), y(a)) \tag{9.19}$$

The proof can be easily extended to piecewise smooth curves. □

We say that a line integral is **independent of path** if, given any path with the same starting and ending points, the integral remains invariant. We say that a region in space is **path-connected** if any two points in the region can be joined by a piecewise-smooth curve lying in the region. We have the following fact:

Theorem 32. *If we have a continuous vector field on an open connected region, then*

$$\int_C \boldsymbol{F} \cdot d\boldsymbol{r} \tag{9.20}$$

is independent of path if and only if $\boldsymbol{F}$ is a conservative vector field.

The proof for this fact will be omitted. Furthermore, we state that a curve C is **closed** if $\mathbf{r}(a) = \mathbf{r}(b)$, which are the starting and ending points of the curve. We have that the following conditions are equivalent; that is, if one is true, then all are true:

1. $\mathbf{F}$ is conservative
2. $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path
3. $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$

The symbol $\oint$ means an integral over a closed curve (or, in the case of surface integrals, a closed surface; but we will see this later).

The **law of conservation of energy** is related to conservative vector fields: if we have a conservative force field, the Hamiltonian (the sum of the potential and kinetic energies) of an object remains constant from point to point. Note that the **potential energy** for a particle in a conservative vector field is defined as $p(x, y, z) = -f(x, y, z)$, where f is the potential function for the vector field. One can use the fundamental theorem of line integrals to derive the law of conservation of energy using basic physics formulas.

We say that a curve C is **simple** if it does not cross itself, and a plane region R is **simply connected** if its boundary consists of one simple closed curve. We now examine **Green's Theorem:**

Theorem 33. *If R is a simply connected region with a piecewise smooth boundary C, oriented counterclockwise, and M and N have continuous partial deriva-*

tives, then

$$\oint_C Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA \tag{9.21}$$

Proof. Suppose, for the sake of simplicity, that R is vertically and horizontally simple. We can split up the boundary curve into two curves C_1 and C_2 defined by $y = f_1(x)$ and $y = f_2(x)$ respectively, so

$$\oint_C Mdx = \int_{C_1} Mdx + \int_{C_2} Mdx \tag{9.22}$$

$$= \int_a^b M(x, f_1(x))dx + \int_b^a M(x, f_2(x))dx \tag{9.23}$$

$$= \int_a^b [M(x, f_1(x)) - M(x, f_2(x))] \tag{9.24}$$

$$= -\int_a^b M(x, y)\Big|_{f_1(x)}^{f_2(x)} dx \tag{9.25}$$

$$= -\int_a^b \int_{f_1(x)}^{f_2(x)} \frac{\partial M}{\partial y} dydx \tag{9.26}$$

$$= -\iint_R \frac{\partial M}{\partial y} dA \tag{9.27}$$

Similarly, we can evaluate $\oint_C Ndy$. Adding the two expressions yields the desired expression. □

A corollary of Green's Theorem is that if we have a plane region R bounded by a a piecewise smooth simple closed curve C oriented counterclockwise, then the area of R can be calculated by the following expression:

$$A = \frac{1}{2} \oint_C xdy - ydx \tag{9.28}$$

We now consider **parametric surfaces**. A parametric surface is the set of points given by the equation

$$\mathbf{r}(u,v) = \langle x(u,v), y(u,v), z(u,v) \rangle \tag{9.29}$$

where $x = x(u,v)$, etc. are called the **parametric equations** for the surface.

We can find a normal vector at the point $(x_0, y_0, z_0) = (x(u_0,v_0), y(u_0,v_0), z(u_0,v_0))$ by calculating

$$\mathbf{N} = \mathbf{r}_u(u_0,v_0) \times \mathbf{r}_v(u_0,v_0) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{vmatrix} \tag{9.30}$$

Note that $\mathbf{r}_u$ and $\mathbf{r}_v$ are vectors tangent to the surface.

The **surface area** of a surface over a region D in the uv-plane is given by

$$\iint_S dS = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\| dA \tag{9.31}$$

A **surface integral** of a function f over a surface S is defined to be

$$\iint_S f(x,y,z)dS = \lim_{\|\Delta\| \to 0} \sum_{i=1}^{n} f(x_i, y_i, z_i)\Delta S_i \tag{9.32}$$

and if the surface takes the form of the equation $z = g(x,y)$, we may be able to calculate it like so:

$$\iint_S f(x,y,z)dS = \iint_R f(x,y,g(x,y))\sqrt{1 + [g_x(x,y)]^2 + [g_y(x,y)]^2} dA \tag{9.33}$$

where R is the projection of S onto the xy-plane.

We say that a surface is **orientable** if we can define a unit normal vector $\mathbf{N}$ continuously over every point on the surface.

There also exists an integral known as the **flux integral**, which, if we have a vector field $\mathbf{F}(x, y, z) = \langle M, N, P\rangle$ with continuous first partial derivatives and a surface S oriented by a unit normal vector $\mathbf{N}$, then the flux integral is

$$\iint_S \mathbf{F} \cdot \mathbf{N} dS \tag{9.34}$$

If the surface S is given by the equation $z = g(x, y)$, and R is the projection of S onto the xy-plane, then we must have that

$$\iint_S \mathbf{F} \cdot \mathbf{N} dS = \iint_R \mathbf{F} \cdot \langle \mp g_x(x, y), \mp g_y(x, y), \pm 1\rangle dA \tag{9.35}$$

depending on if S is oriented upward (the top sign) or downward (the bottom sign).

The flux integral allows us to state **Gauss's law**:

$$\oiint_S \mathbf{E} \cdot \mathbf{N} dS = 4\pi q \tag{9.36}$$

where $\mathbf{E}$ is an electric field, S is a closed surface (such as a sphere), and q is the total charge enclosed inside the surface.

We now consider one of the most important theorems in vector calculus, the **Divergence Theorem**:

Theorem 34. *Suppose that we have a solid region Q bounded by a closed and orientable surface S. We must have that*

$$\oiint_S \boldsymbol{F} \cdot \boldsymbol{N} dS = \iiint_Q \nabla \cdot \boldsymbol{F} dV \tag{9.37}$$

Proof. Suppose $\mathbf{F}(x, y, z) = \langle M, N, P\rangle$. Then we have

$$\oiint_S \mathbf{F} \cdot \mathbf{N} dS = \oiint_S \langle M, N, P\rangle \cdot \mathbf{N} dS \tag{9.38}$$

We consider the equation

$$\oiint_S \langle 0,0,P\rangle \cdot \mathbf{N} dS \tag{9.39}$$

Suppose that we have a **simple solid** region with an upper surface

$$z = g_2(x,y) \tag{9.40}$$

and a lower surface

$$z = g_1(x,y) \tag{9.41}$$

such that R is the projection of both onto the xy-plane. If there exists an additional lateral surface, the normal vector must be horizontal, and $\langle 0,0,P\rangle \cdot \mathbf{N}$ will vanish. Therefore, we only consider the upper and lower surfaces:

$$\oiint_S \langle 0,0,P\rangle \cdot \mathbf{N} dS = \iint_{S_1} \langle 0,0,P\rangle \cdot \mathbf{N} dS + \iint_{S_2} \langle 0,0,P\rangle \cdot \mathbf{N} dS \tag{9.42}$$

One can verify visually that

$$\iint_{S_1} \langle 0,0,P\rangle \cdot \mathbf{N} dS = \iint_R \langle 0,0,P(x,y,g_1(x,y))\rangle \cdot \left\langle \frac{\partial g_1}{\partial x}, \frac{\partial g_1}{\partial y}, -1 \right\rangle dA \tag{9.43}$$

$$= -\iint_R P(x,y,g_1(x,y)) dA \tag{9.44}$$

and likewise,

$$\iint_{S_2} \langle 0,0,P\rangle \cdot \mathbf{N} dS = \iint_R P(x,y,g_2(x,y)) dA \tag{9.45}$$

Hence, adding the two expressions, we obtain

$$\oiint_S \langle 0,0,P\rangle \cdot \mathbf{N} dS = \iint_R [P(x,y,g_2(x,y)) - P(x,y,g_1(x,y))] dA \tag{9.46}$$

$$= \iint_R \left[\int_{g_1(x,y)}^{g_2(x,y)} \frac{\partial P}{\partial z} \right] dA = \iiint_Q \frac{\partial P}{\partial z} dV \tag{9.47}$$

Similarly, for M and N, we have

$$\oiint_S \langle M,0,0\rangle \cdot \mathbf{N} dS = \iiint_Q \frac{\partial M}{\partial x} dV \tag{9.48}$$

and

$$\oiint_S \langle 0,N,0\rangle \cdot \mathbf{N} = \iiint_Q \frac{\partial N}{\partial y} dV \tag{9.49}$$

Adding the three together, we obtain

$$\oiint_S \langle M,N,P\rangle \cdot \mathbf{N} dS = \iiint_Q \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z}\right) dV \tag{9.50}$$

$$= \iiint_Q \nabla \cdot \mathbf{F} dV \tag{9.51}$$

This gives us the desired expression, proving the Divergence Theorem. □

We can classify points in a vector field like so:

1. **Source**: $\nabla \cdot \mathbf{F} > 0$ at the point
2. **Sink**: $\nabla \cdot \mathbf{F} < 0$ at the point
3. **Incompressible**: $\nabla \cdot \mathbf{F} = 0$ at the point

We also have **Stokes's Theorem**, not to be confused with Stokes's Theorem on manifolds:

Theorem 35. *If S is an orientable surface with a unit normal vector $\boldsymbol{N}$ bounded by a piecewise smooth simple closed curve C, then*

$$\oint_C \boldsymbol{F} \cdot d\boldsymbol{r} = \iint_S (\nabla \times \boldsymbol{F}) \cdot \boldsymbol{N} dS \tag{9.52}$$

The proof will be omitted.

We define the **circulation** of a vector field $\mathbf{F}$ around a circle C_α to be

$$\oint_{C_\alpha} \mathbf{F} \cdot \mathbf{T} ds \tag{9.53}$$

where $\mathbf{T}$ (or $\mathfrak{t}$ I guess) is the unit tangent of the curve. If α is the radius of C_α, then we can use Stokes's Theorem to show the curl of $\mathbf{F}$ can be thought of in this manner:

$$(\nabla \times \mathbf{F}) \cdot \mathbf{N} = \lim_{\alpha \to 0} \frac{1}{\pi \alpha^2} \oint_{C_\alpha} \mathbf{F} \cdot \mathfrak{t} ds \tag{9.54}$$

This expression is known as the **rotation** of $\mathbf{F}$ about $\mathbf{N}$. This concludes our brief excursion into vector calculus. Now we will unify the theorems we have learned through the abstract language of differential forms, which are not usually taught in a multivariable calculus class.

Chapter 10

Differential Forms and Stokes's Theorem